

LINUX FUNDAMENTALS AND TERMINAL

**SINGLE-USER OPERATING
SYSTEMS**

Javier Ramos Pérez

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1. Linux distributions

It is a set of programs and files that are prepared to be installed on the system. Linux has several families and distributions created from it, such as:
RedHat–Fedora, SuSe, Debian, Linux Mint, Ubuntu, etc.

1.1. The linux terminal (shell)

It is a program that allows the user to execute commands interpreted by it.
The most commonly used versions are sh and bash. To check which one we are using, we use the command `echo $SHELL`.
It has an integrated command history using the cursor arrows and `.bash_history`.
Autocomplete is available with the TAB key.
It is case-sensitive.

- When started, 6 consoles are created in text mode and one in graphical mode. To access these, we can:
- From the graphical environment (Ctrl+Alt+T).
- Using a key combination: (Ctrl+Alt+F3/F4/F5.../F7).
- To return to the graphical interface: Ctrl+Alt+F2.
- Switch between them with Alt+Fx.

1.2. Linux users

- **Common users:** Restricted access to certain files and resources, such as other users' files.
- **System users:** Cannot log in with these; they are only used to start processes with required permissions.
- **'root' user (superuser):** Full access to the entire system.

4. The directory tree

Using the `tree` command in the terminal, we can see all the directories with their contents in a descending tree structure.

5. File system structure

All of these are located in the root directory (/):

- `/bin/` → Essential commands.

- **/boot/** → Boot files.
- **/dev/** → System devices.
- **/etc/** → Configuration.
- **/home/** → User directories.
- **/lib/** → Libraries and kernel modules.
- **/media/** → Mounting of removable devices.
- **/mnt/** → Temporary mount point for file systems.
- **/opt/** → Additional software.
- **/sbin/** → System binaries.
- **/srv/** → Service data.
- **/tmp/** → Temporary files.
- **/usr/** → Applications and utilities.
- **/var/** → Variable files.
- **/root/** → Superuser's folder.
- **/proc/** → Kernel and process information.

6. Absolute and relative path

6.1 Absolute path

- A file or directory referenced from the root directory to the desired location, being unique.
Example: /home/alumnos/Desktop/SOM/Unit4.

6.2 Relative path

- A file or directory referenced from the current directory to the desired location, using "." to move back directories.
Example: ../SOM/Unit4.

7. File and folder properties

- Each file and folder has permissions that can be viewed using the `ls -l` command.

- The first character indicates the file type:
 - - → Normal file.
 - d → Directory.
 - l → Symbolic link.
 - c → Character device file.
 - b → Block device file.
- The groups of three characters represent the permissions for different types of users:
 - r → read
 - w → write
 - x → execute
 - - → no permission
- • The first group of three characters is for the owner's permissions.
- • The second group is for the permissions of the users in the group.
- • The last group is for the permissions of other users.

8. I-NODE

- •Data structure where file information is stored, such as:
 - i-node number.
 - File type.
 - User and group.
 - Permissions or access modes.
 - Creation, access, and modification dates.
 - Links.
- Each group of three characters can be read as a "binary string," with a '1' if the permission is enabled and a '0' if not.
- This binary string is generally converted into the octal system to represent all permissions (owner, group, others).

Example: r-x → 101 rwx r-x r-- → 754

9. File / Directory

- A set of information related to the same concept under an identifying name.

- **Filename**
 - Between 1 and 255 characters.
 - The '/' character cannot be used.
 - Case-sensitive.

10. Environment variables

- Labels that store values.
- Their scope is the active session or command prompt.
- Create variable: `variable_name=variable_value`.
- Get the value of the variable: `$variable_name`.

10.1. Predefined environment variables

- Some variables are predefined in the system.
- You can use the `set` command to display the full list.
- Some of the most important ones are:
 - `$DISPLAY` → Characteristics of the graphical environment.
 - `$HOME` → Current user's working directory.
 - `$HOSTNAME` → Hostname
 - `$PATH` → Locations where the shell will search for executable files.
 - `$PS1` → Session prompt
 - `$PWD` → Current directory.
 - `$SHELL` → Version of the shell in use.
 - `$UID` → ID of the logged-in user.
 - `$USER` → Logged-in username.

11. System boot

The Linux boot process is a series of stages that occur when a Linux system is powered on.

1. **BIOS/UEFI Initialization:** When powered on, the BIOS/UEFI performs a hardware check and looks for a boot device (hard drive or USB).

2. **Bootloader (GRUB):** Loads the kernel into memory and shows a menu to choose the operating system or kernel to boot.
3. **Kernel Loading:** The kernel is loaded into memory, initializes the hardware, mounts a temporary filesystem, and then mounts the root filesystem.
4. **Init System (systemd):** Manages services and sets up the system environment.
5. **Target and Service Initialization:** systemd activates services such as networking, logging, and user sessions in parallel for greater efficiency.
6. **Login Prompt:** Displays a login prompt on the console or a graphical login screen to access the system.

11.1. System targets

- **poweroff.target:** Shutdown the system.
- **rescue.target:** Single-user mode (Rescue).
- **multi-user.target:** Multi-user mode without networking (level 2), multi-user mode with networking (level 3), customizable multi-user mode (unused, level 4).
- **graphical.target:** Multi-user mode with networking and GUI.
- **reboot.target:** Reboots the system.

12. Configuration files

Some of the most important configuration files are:

- **/boot/grub/grub.cfg** → Boot loader configuration.
- **/etc/hosts** → Resolution of local PC names to IP.
- **/etc/fstab** → List of the existing file systems on the computer.
- **/etc/profile** → Shell script that runs when any user logs in.

Within **/etc**, there are also configuration files for system users and groups. Examples include:

- **/etc/passwd** → Information about system users.
- **/etc/shadow** → Encrypted user passwords.

- **/etc/group** → Existing groups on the system

12.1. Configuration files

/boot/grub/grub.cfg: Configures system boot with GRUB. It is generated using **/etc/default/grub** and allows you to change the boot time and default operating system. After modifying it, the command **update-grub** must be run.

- **/etc/fstab:** Lists the file systems and how they should be mounted when the system starts, including the device, mount point, file system type, and mount options.
- **/etc/profile:** Script ejecutado al iniciar sesión, configurando parámetros para todos los usuarios. No se puede modificar por usuarios comunes.
- **\$HOME/.profile:** Script executed at login, setting parameters for all users. It cannot be modified by regular users.
- **\$HOME/.bash_logout:** Script executed after **/etc/profile** at login. The user can modify it.
- **\$HOME/.bash_history:** Stores the last commands executed by the user.